

Your grade: 100%

Next item →

1. Tomasi-Kanade factorization is particularly useful in Structure from Motion(SfM) when dealing with:

1 / 1 point

- ☐ Scenes with a single image and one camera.
- ☐ Large-scale outdoor scenes with moving cameras.
- ☒ Multiple views of a 3D scene with feature point correspondences.
- ☐ Stereo vision problems with known baseline distances.

✔ Correct

Tomasi-Kanade factorization is most useful in SfM when dealing with multiple views of a 3D scene, where you have correspondences between feature points across these views. This is because the method aims to recover the 3D structure of a scene and camera motion from such data.

2. When applying SVD in Tomasi-Kanade factorization for 3D reconstruction, what information is extracted from the singular value matrix?

1 / 1 point

- ☐ Rotation matrix
- ☐ Translation Matrix
- ☐ Intrinsic and Extrinsic camera parameters
- ☒ Motion Matrix and Structure Matrix

✔ Correct

Using SVD,

$$W=U\Sigma V^T$$

$$M \text{ (Motion Matrix)} =U(\Sigma)^{1/2}Q$$

$$S \text{ (Structure Matrix)} =Q^{-1}(\Sigma)^{1/2}V^T$$

Q can be evaluated by enforcing the orthonormality constraint.

3. Which of the following best describes the primary input data required for Tomasi-Kanade factorization in SfM?

1 / 1 point

- ☒ 2D image keypoints and camera poses
- ☐ 3D point cloud and camera calibration parameters
- ☐ Silhouette masks of objects in the scene
- ☐ Depth maps and image gradients

✔ Correct

The primary input data for Tomasi-Kanade factorization in SfM typically includes 2D image keypoints (feature points) and camera poses. These keypoints are tracked across multiple images, and camera poses represent the positions and orientations of cameras at different time instances.

4. In the context of SfM using Tomasi-Kanade, what is the typical output of the factorization process?

1 / 1 point

- ☐ Depth maps for each frame in a video sequence.
- ☒ Camera trajectories and 3D point cloud.
- ☐ Motion blur estimates for each frame.
- ☐ Camera focal length adjustments.

✔ Correct

Using Tomasi-Kanade factorization, the typical output of the factorization process includes estimates of camera trajectories (camera motion) and a 3D point cloud representing the 3D structure of the scene. The method aims to recover both camera motion and scene structure from the input data.